



SIT Meeting #4

Programming Languages

Presented by Vice President Mr. Zahidi



What is a Programming Language?

To put it simply, a programming language is a language one can use to communicate with a computer in order to get it to do what you want. Just like how you would request a waiter to receive an order, you can treat a computer as your digital waiter.



How does it work?

Let us take the analogy of a waiter.

You want to order a specific meal, a task, that you have in your mind.

Let's say you want sushi:



How does it work?



In order to get your desired sushi,
you must tell the waiter to tell the
chef to make sushi.

The programming language is the
way to talk to the “binary chef” in
order to get your desired “meal”



Levels of Programming Languages

How do we decide what to tell the waiter?

First we consider the level of control we want,
the level of the programming language.



Programming languages are sorted into
“levels”.

High-level, low-level, and
“assembly-languages” (a mix of both)

High-level Languages

A high-level language is like telling your waiter what you want from a pre-made menu.

Very efficient, but you lack a lot of fine control for what you want.



High-level languages include ones such as Python, Java, C/C++, Javascript, and most other languages you may know of.

Low-level Languages

A low-level language is like talking directly to the chef, going into intricate detail to control every single part of the cooking process, down to the last step; the temperature adjustment, the exact grammes of spices used, etc.



Low-level languages are specific to each piece of hardware, and take up less memory. However, they are notoriously difficult to use.

Also referred to as machine code.

Assembly Languages

```
MONITOR FOR 6802 1.4          9-14-80  TSC ASSEMBLER  PAGE    2

C000                ORG      ROM+$0000 BEGIN MONITOR
C000 8E 00 70  START  LDS      #STACK

*****
* FUNCTION: INITA - Initialize ACIA
* INPUT: none
* OUTPUT: none
* CALLS: none
* DESTROYS: acc A

0013      RESETA EQU    %00010011
0011      CTLREG EQU    %00010001

C003 86 13      INITA  LDA A  #RESETA  RESET ACIA
C005 87 80 04      STA A  ACIA
C008 86 11      LDA A  #CTLREG  SET 8 BITS AND 2 STOP
C00A B7 80 04      STA A  ACIA

C00D 7E C0 F1      JMP      SIGNON    GO TO START OF MONITOR

*****
* FUNCTION: INCH - Input character
* INPUT: none
* OUTPUT: char in acc A
* DESTROYS: acc A
* CALLS: none
* DESCRIPTION: Gets 1 character from terminal

C010 B6 80 04  INCH  LDA A  ACIA      GET STATUS
C013 47          ASR A      SHIFT RDRF FLAG INTO CARRY
C014 24 FA      BCC  INCH  RECIEVE NOT READY
C016 B6 80 05      LDA A  ACIA+1    GET CHAR
C019 84 7F      AND A  #$7F      MASK PARITY
C01B 7E C0 79      JMP      OUTCH   ECHO & RTS

*****
* FUNCTION: INHEX - INPUT HEX DIGIT
* INPUT: none
* OUTPUT: Digit in acc A
* CALLS: INCH
* DESTROYS: acc A
* Returns to monitor if not HEX input

C01E 8D F0      INHEX BSR  INCH  GET A CHAR
C020 81 30      CMP A  #'0      ZERO
C022 2B 11      BMI  HEXERR  NOT HEX
C024 81 39      CMP A  #'9      NINE
C026 2F 0A      BLE  HEXRTS  GOOD HEX
C028 81 41      CMP A  #'A
C02A 2B 09      BMI  HEXERR  NOT HEX
C02C 81 46      CMP A  #'F
C02E 2E 05      BGT  HEXERR
C030 80 07      SUB A  #7      FIX A-F
C032 84 0F      HEXRTS AND A  #$0F  CONVERT ASCII TO DIGIT
C034 39          RTS

C035 7E C0 AF  HEXERR JMP      CTRL    RETURN TO CONTROL LOOP
```

An Assembly language is a sort of in-between between high-level and low-level. Instead of picking from a menu or controlling every part of a process, it is like telling the waiter what specific food items you want.

Assembly languages are second-generation, and provide a layer of abstraction on top of low-level languages, though are not as abstract as high-level languages.

Language Paradigms

A language paradigm is like a style of a language.

The two main paradigms you may be familiar with are “procedural” and “object-oriented”

Procedural languages are organised around functions and steps, whereas object-oriented languages are far more scalable, as one can make representations of simulated or real-world objects for multiple uses.



Language Paradigms

OOP languages include:
C#, Java, and Multi-Paradigm
languages

Procedural languages include:
C, BASIC, Pascal



yay ^_^ wahoo!!
yippee :D
yahoooooooo >_<



That's it. Thank you.

honestly quite incredible
honestly quite incredible

honestly quite incredible

honestly quite incredible

